NUIMPACT

Sector Spotlight Digest

Fall 2025

Northeastern University

NUIMPACT
IMPACT INVESTING INITIATIVE

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NUImpact Overview

About NUImpact

- NUImpact is Northeastern University's student-led impact investing initiative and investment fund.
- Since our inception in Spring 2016, the organization has served as a unique resource and thought exchange for the Northeastern community to understand purposeful capital, develop technical skills, and make targeted investments in companies that promote positive social and environmental impact
- Our three-pronged investment thesis seeks investment in companies that are:

Northeast-Focused | **Aid Underserved Communities** | **Financially Sustainable**

Our Research Arm Yields Significant Benefits

- The research analyst program places a strong emphasis on industry research to enhance our fund's impact diligence process.
- A key responsibility for research analysts is crafting concise one-page reports spotlighting a chosen sector within their verticals.
- Some benefits of this new program include:

Informed Decision Making

Offers the fund with valuable insights into specific industries, enabling risk mitigation and better-informed investment decisions

Refined Impact Assessment

Enhanced sector understanding equips analysts with knowledge needed to conduct robust impact diligence and measurement

Industry Focus

- **Education & Media**
- **Energy & Environment**
- **Finance & Technology**
- **Food & Agriculture**
- **Healthcare**

Meet the Research Team

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Education & Media: EdTech and Online Tutoring

Industry Overview

Defining Sector

Definition: Online tutoring services use digital platforms to deliver educational tools, allowing students to learn

K12 Education Apps	Test Preparation Platforms	MOOC Portals
Tech Learning Solutions	Gamified Learning Apps	Language Learning Platforms
Career Development Apps	Learning Management Systems	School Administration

Impact Overview

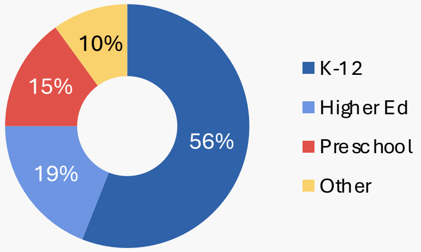
- **Bridging educational gaps:** Provides educational access for underserved communities through exam preparation support (SAT, ACT, GRE, GMAT) and improved language accessibility
- **Flexibility and accessibility:** Removes geographical and logistical barriers to learning

The growth of online tutoring reflects a shift toward technology-driven education, positioning digital learning tools as essential for accessibility to learning

Trends

- Rising demand for skills in coding and data science, STEM courses amount for 16.1% CAGR. STEM captures 53%+ of the K-12 Edtech sector[7]
- EdTech market include Learning Management Systems, Language Learning Platforms, STEM and Coding Platforms, Adaptive Learning tech, etc.[8]

Global EdTech Market Distribution[2]



Market Growth

- EdTech Sector is \$10.4-10.9B in 2024 and projected to grow to \$22.7-23.7B by 2030[4]
- Rapid growth in EdTech industry due to developing technology and dependency on technology.
- Government initiatives and increased public education budgets for digital infrastructure

\$10.7B
2024

\$23.2B[1]
2030

Headwinds

Systemic socio-economic gaps limit access to tactic online learning and digital literacy development

Privacy and transparency concerns remain around data systems and algorithms used in education

Increasing digitalization raises ethical concerns about reduced human interaction in learning[2]

Tailwinds

Hybrid and flexible learning models have become a prominent feature of education systems[3]

Growing openness to digitalization among teachers, parents, and institutions[2]

Rapid advancements in AI are enhancing personalized learning[3]

Recent Deal Highlights

popple

- Raised \$25,000 in seed funding through BU's Innovation Pathway[5]
- Selected for 2024 MassChallenge Boston accelerator (youngest team in cohort)
- Reached Hult Prize Global Summit in Lisbon (competed among 20,000 students)

GROWBIE

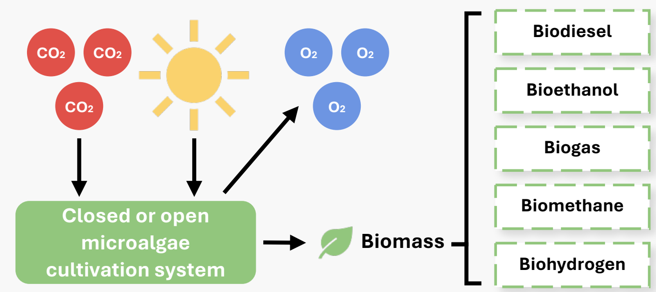
- Received non-equity funding from StartOut (2024)
- Reached 2,000 paying customers with \$0 spent on marketing within first year of launch[6]
- Startup Boston's Student Founder of the Year 2024
- Listed as one of Boston's "Startups to Watch in 2024" by BostInnovation

Energy & Environment: Algae Carbon Solutions

Industry Overview[1]

- **Algae Carbon Solutions** use microalgae grown in photobioreactors to capture industrial CO_2
- Natural land-based systems close the loop between **carbon capture + product generation**, advancing **both climate and economic returns**
- Algae can capture **10-50 x more CO_2 per acre** than terrestrial photosynthesizing plants
- Projected market size to reach **$7.3 Billion by 2030 (2030: CAGR 36%)**

Process Overview



Tailwinds & Market Growth[3]

Policy and Investment Momentum[2]

- **U.S. DOE Bioenergy Technologies** funding microalgae R&D for carbon-negative manufacturing
- **Inflation Reduction Act §45Q** subsidizes CO2 utilization projects, supporting early deployments of algae-based bioproducts
- **$350 Million VC and Corporate Carbon Tech** investments since 2023

Technological Advancements

- **Photobioreactor efficiency:** Increased by 30% with AI-Optimized light and CO_2 injection

Industry Headwinds[4]

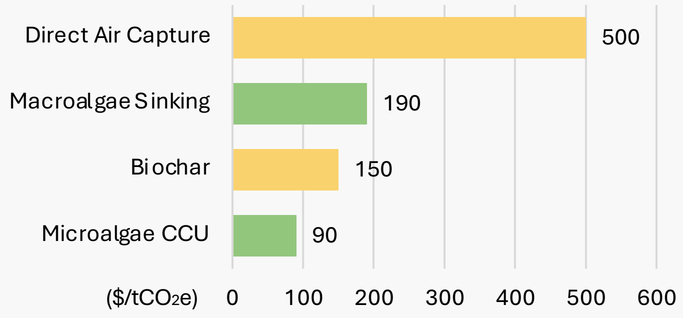
High CapEx:
Initial photobioreactor investment costs around ($5-10 Million per facility)

Energy Balance:
Pumping & Drying may affect total carbon abatement without renewable integration

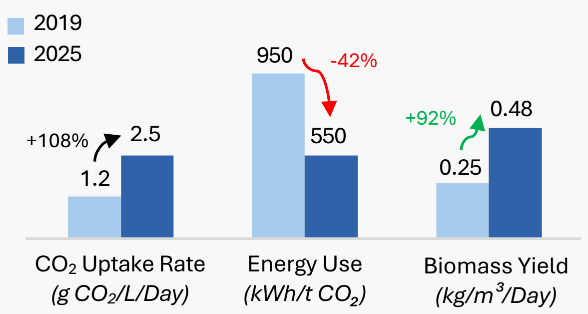
Policy & Credit Uncertainty:
Risks surrounding IRA §45Q incentives and fluctuating carbon prices make projects volatile

Current State[5]

Cost per Metric Ton of CO_2 removed, in U.S. dollars per ton ($/tCO$_2$e)



Performance Improvements in Land-Based Algae Carbon Solutions (2019-2025)



Recent Deal Highlights

CREW Carbon (New Haven, CT)[6]

- **$32.1 million offtake agreement** secured with Frontier Climate to remove ~71,900 tons of CO_2 between 2025-2030
- Operates a **bio-mineralization platform** that turns captured CO_2 into solid carbonates

Clean Corp Technologies (Holyoke, MA)[7]

- **$1.2 million from Massachusetts' M2I2 program** awarded to scale its plasm-based contaminant removal systems for agricultural applications
- Incorporates **low-energy, non-thermal gas processing** for enhanced food safety measures

Finance & Technology: Digital Assets and Currencies

Industry Overview

Digital Assets and currencies in the Fintech market in the United States[3]

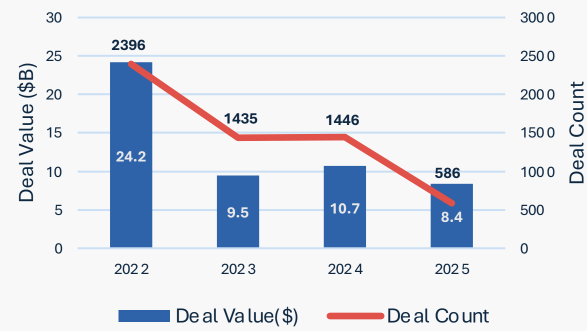
$18.6B in 2025 → **CAGR of 3.53%** → **$19.3B by 2026**

166.7M — The US is expected to have 166.7M **Users of Digital Assets** by 2026, with a 47.76% penetration rate[4]

18% — 18% of publicly traded companies in North America now hold digital assets on their balance sheets

Trends

Total global funding activity (VC, PE, and M&A) in digital assets and currencies (2022-2025)[1]



Future Outlook

Tailwinds[5]

The **change in the US administration** drives recognition towards digital assets and currencies and drives major impacts worldwide[1]

The repeal of **SAB 121** enables banks to **hold crypto without heavy capital limits**, spurring institutional investment and **growth in U.S. digital assets**[2]

The US introduced the **GENIUS Act stablecoin bill**, recently passed by Congress, which provides a regulated framework and guardrails for stablecoins[7]

Headwinds[5]

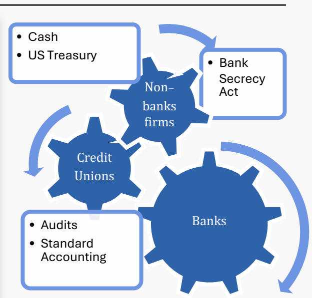
Regulatory ambiguity U.S. law on digital assets with lack of a **clear legal category** for "cryptocurrency"

Inconsistent or evolving **accounting, reporting and supervisory standards** increase compliance difficulties

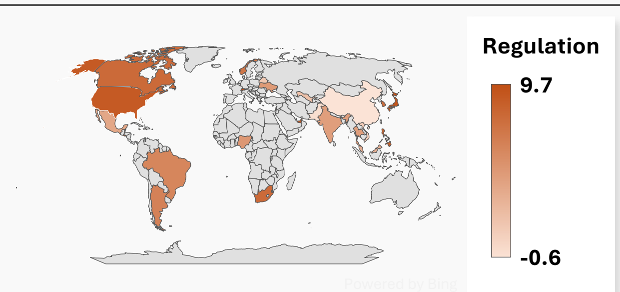
U.S.-based digital-asset companies face an **uneven playing field compared to other jurisdictions**, which reduces U.S. competitiveness in innovation

Focus on the GENIUS Act

Only **approved financial institutions** can issue stablecoins under **Federal Reserve oversight**, **maintaining 1:1 reserves** in safe assets and complying with **Bank Secrecy Act rules** to ensure transparency and consumer protection.



Crypto Asset Regulation in 2025[6]



Recent Deal Highlights

CIRCLE

- Circle's IPO raised over $1 billion, positioning its stablecoin ecosystem for mainstream finance and financial inclusion[8]
- In 2024, **$161 billion in remittances** flowed from the U.S. to Latin America and the Caribbean. Average transfer cost **remains 5.8%,** heavily impacting **low-income senders** in the **Northeast U.S.**
- Circle's USDC network **lowers fees** and enables **faster**, smaller-value

ripple

- Ripple's blockchain **speeds up cross-border payments** and **cuts costs**.
- Together, they make remittances **more inclusive**, **affordable**, and **equitable**, helping marginalized communities keep more of their money

Food & Agriculture: CEA Systems

Sector Overview

Industry Overview [1]

- Controlled Environment Agriculture (CEA) grows crops in controlled spaces using **hydroponic, aeroponic, or aquaponic** systems for production
- High-Tech Greenhouses: combines sunlight with LED lighting for energy-efficient growth
- Vertical Farms: rely on LEDs entirely for maximum control and yield in limited space

CAGR: 18.1%[7]

$96.1B 2024 → **$507.3B** 2034

Impact Overview [1]

CEA vs Traditional Farming: Key Performance Indicators

Metrics	CEA Advantage vs. Trad. Farming	Methods
Water Use	85-90% less water through recirculation	Hydroponics, Aeroponics
Yield	7-35x higher annual crop output	Hydroponics, Vertical Farms
Energy Use	30-120x more energy (LEDs, HVAC)	All CEA Systems
Env. Impact	Lower fertilizer runoff, shorter supply chains	All CEA Systems

In New England, short seasons and limited farmland, CEA methods are ideal: cuts transport emissions, conserves up to 90% of water; enabling higher yields

Future Outlook

Tailwinds

Mass Save and NYSERDA's Agricultural Energy Efficiency Program are accelerating investment in energy-efficient greenhouses and indoor farming[4]

Increasing global demand for food: approx. 60% more food will be needed to feed a population of 9.3b people, projected to exceed within 30 years

Technological advancement in LED lighting, sensors, and automation is lowering operating costs and improving yield consistency

Headwinds

High energy and maintenance costs make profitability difficult for early stage or smaller farms

Large upfront capital requirements and long payback periods limit access to financing

Economic viability remains concentrated in leafy green and herbs, limited progress on fruiting crops

CEA Methods[5]

Hydroponic

Plants grow in nutrient-rich water without soil

Pros
- High yield, efficient
- Up to 90% less water

Cons
- Energy intensive
- Constant monitoring

Aeroponic

Roots suspended in air and misted with nutrient solution

Pros
- Boosts oxygen and nutrient absorption
- Minimal water usage

Cons
- Technical setup
- Nozzle clogging risk

Aquaponic

Combines fish farming and plant cultivation in closed loop system

Pros
- Reduces fertilizer costs with fish waste recycling

Cons
- Limited scalability
- Hard to balance ecosystems

Recent Deal Highlights

Re-Nuble

- Converts food waste into organic liquid nutrients for hydroponic and soil less farming[3]
- Raised $1.9M+ to date, backed by Black Founders Matter, Unreasonable Group, and USDA initiatives[2]
- Helps farms lower costs and boost yields with peat-free, sustainable nutrient systems

We Grow Microgreens

- Urban hydroponic farm in Hyde Park, Boston, producing microgreens for local restaurants and markets[6]
- Received $134k City of Boston CPA Grant and USDA support for sustainable infrastructure
- Promotes local food access and community based urban agriculture

Healthcare: Maternal Health Equity

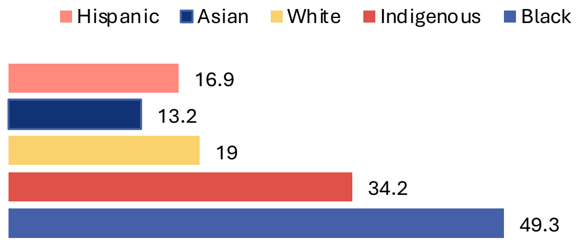
Industry Overview

- **Maternal health equity** means ensuring all women and birthing people have **fair access to quality care** during **pregnancy, childbirth, and postpartum**
- In Massachusetts, Black women are twice as likely as white women to die from pregnancy-related causes[2]
- Focus areas include expanding Medicaid postpartum coverage, improving culturally competent care, and reducing hospital-based disparities

22.3
Deaths per 100k

U.S. national maternal mortality rate of per 100,000 live births: the highest among developed countries[1]

Maternal Mortality Rate by Race (U.S., 2022)



Headwinds

Hospital Variation: Within major cities like Boston and New York, outcomes differ between some hospitals serving wealthier vs. lower-income populations[2]

Rural Shortage: Rural parts of Maine, Vermont, and upstate New York have experienced OB-GYN shortages, forcing patients to travel long distances for care[3]

Federal Funding Cuts: 24 million women rely on Medicaid for maternal health and policies being considered would dramatically increase maternal mortality[4]

Tailwinds

Medicaid Expansion: All Northeastern states have extended postpartum Medicaid coverage to 12 months, ensuring continuity of care for low-income mothers[7]

Policy Innovation: States like Massachusetts and New York have launched maternal mortality review committees and equity dashboards to track racial gaps in outcomes[11]

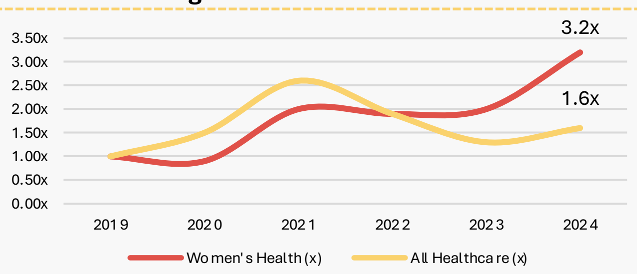
Academic Research Hubs: Northeastern universities (Harvard, Columbia, Yale) are pioneering studies on maternal mental health, racial bias in care, and AI-driven risk prediction models for complications[5]

Investment Trends

Women's Healthcare is no Longer Niche

- Total VC dollars going to women's health **more than tripled** between 2019 and 2024, outpacing healthcare as a whole
- Biopharma in women's health is attracting investment the most
- Ongoing research is helping to identify more areas of women's health that can benefit from new drugs and treatments; investors are catching on[8]

Change in Investment Since 2019



Recent Deal Highlights

Cayaba Care

- Focused on closing racial equity gaps by integrating community health workers into prenatal and maternity care[9]
- Received a **$250,000** investment from the Penn Medicine-Wharton Fund for Health to scale a maternal-health startup

She Matters Alliance

- **She Matters** is a startup focused on improving maternal health outcomes for black mothers and recently
- Raised **$2 million** in funding to address Black maternal morbidity and build a digital platform focused on high-risk minority communities[10]

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